Imperial College London

Lecture 3

CMOS Transistors & the Fabrication Process

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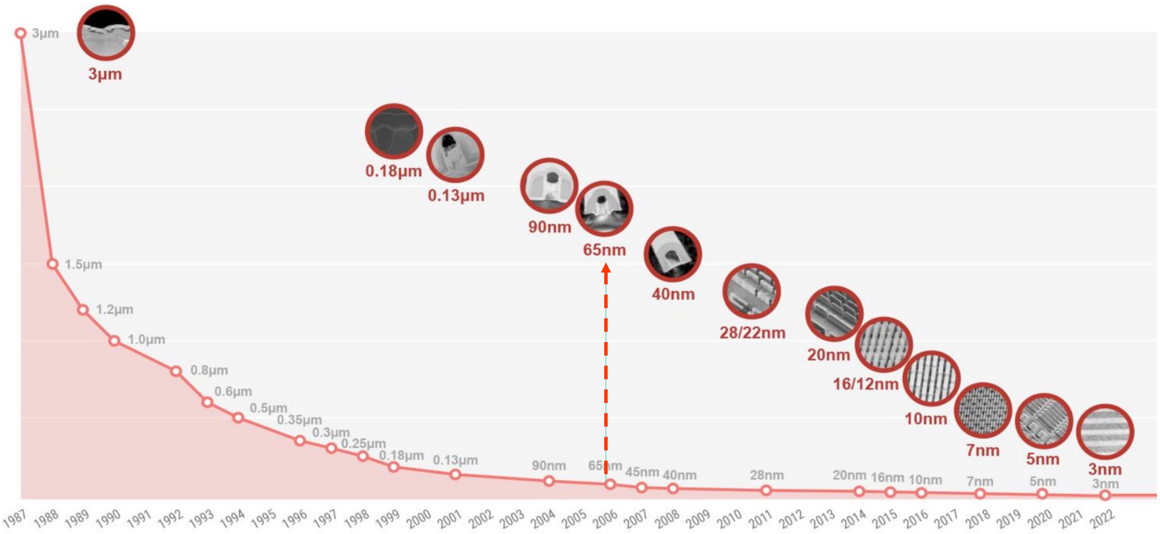
This lecture will cover the following topics:

- A revision on how a MOSFET works
- The characteristics of an ideal NMOS transistor
- The non-ideal effects on an NMOS transistor
- The CMOS fabrication process, step-by-step

This lecture lays the foundation for the understanding of the layout of a VLSI chip.

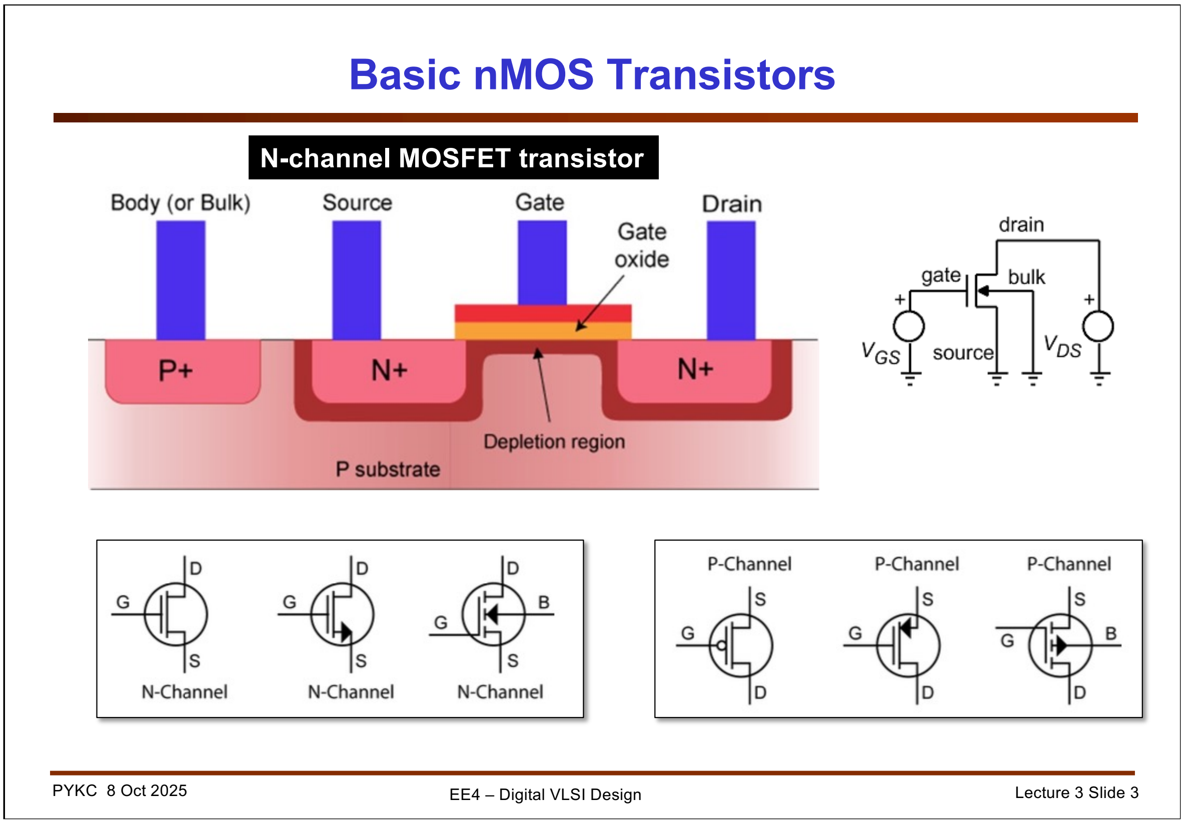
TSMC CMOS Technology



Source: TSMC

This graphs shows the evolution of TSMC's CMOS process over the past four decades. The process we use is the 65nm, which is already nearly 20 years old (but affordable). Highlighted at various technology node is a small photo of the innovation TSMC introduced at each stage to the process improvement.

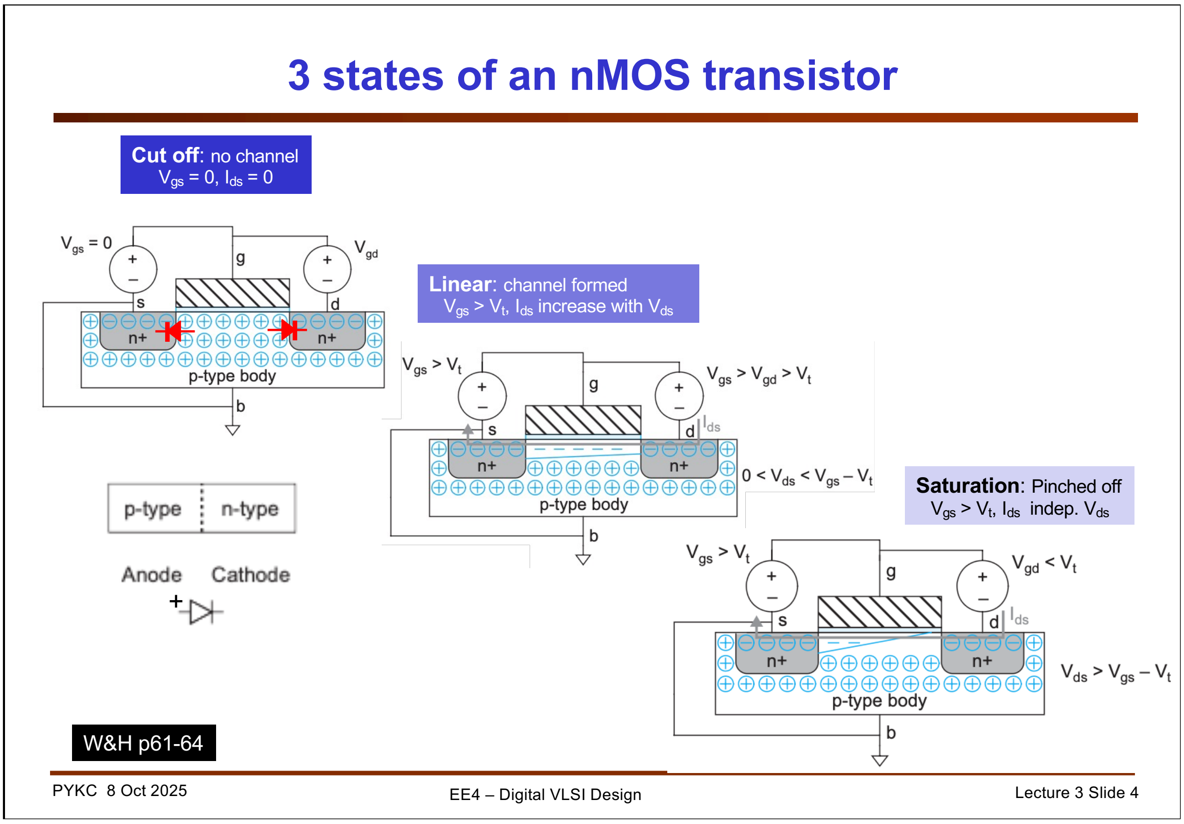
Node	Key Innovation(s)
130 nm	Copper + low-k interconnect, STI, salicide
90 nm	Immersion lithography, improved low-k, CMP scaling
65 nm	Strain enhancement, double patterning, tighter metal/via scaling
45/32 nm	High-k / metal gate, multiple patterning, stronger DFM, variation control
28 nm → 16/12 nm	FinFET adoption, better interconnect, low-power techniques
10 / 7 nm	EUV introduction, aggressive scaling, BEOL innovations
5 / 3 nm	More EUV, scaling of fins/contacts/vias, initial transition to GAA, packaging co-optimization
2 nm / beyond	GAA / nanosheet / CFET, backside power, 3D integration, advanced materials, packaging synergy



- MOS transistors are majority-carrier devices.
- For n-channel transistors, the majority carriers are electrons conducted through a channel.
- A positive gate voltage (w.r.t. substrate) ***enhances*** the number of carriers in the channel and increases conduction.
- **Threshold voltage** V_{tn} denotes the gate-to-source voltage above which conduction occurs.
- For **enhancement** mode devices, V_{tn} is positive; for **depletion** mode devices, V_{tn} is negative. Only enhancement mode devices are used in modern VLSI chips.
- p-channel devices are similar to n-channel devices, except that all voltages and currents are in opposite polarity.

Shown here is the cross-section of an n-channel enhancement transistor:

- Substrate is moderately doped with p-type material. Substrate in digital circuit is usually connected to V_{Gnd} (ground).
- The source and drain regions are heavily doped with n-type material through diffusion. These are often referred to as the **diffusion** regions.



A MOSFET (metal-oxide-silicon field effect transistor) can be n-type or p-type. The slide here shows the working of an n-type or NMOS transistor. It operates in three states:

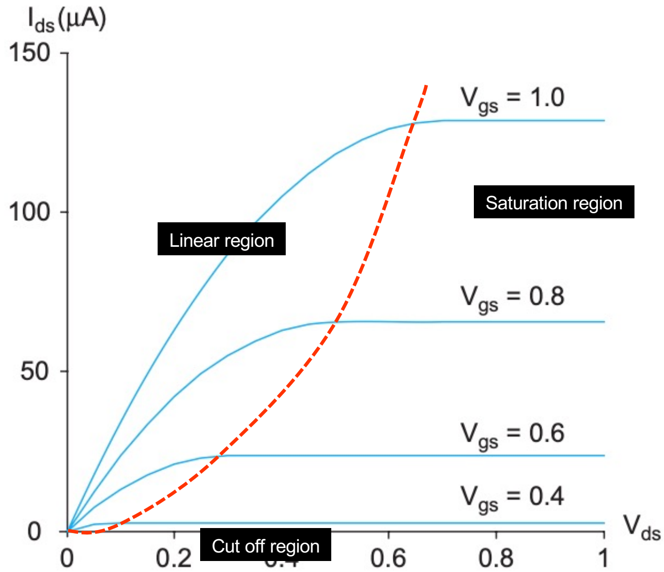
Cut off : When the source terminal is connected to GND and $V_{gs} = 0$, the transistor is in OFF state. The p-n junctions between substrate (body or bulk) and the drain and source form two diodes that are not conducting. Therefore, no current flows between the source to drain.

Linear: If the drain voltage is not low or close to 0V, as the gate voltage increase such that V_{gs} is above the threshold voltage V_t, ($V_{gs} > V_t$)the positive gate voltage repels the major carriers (+ve) in the p-type body and attracts electrons toward the layer under the gate. This connects the n+ regions from source to drain, forming a **channel** through which current can flow. Under the gate, what used to be p-type now become n-type material. This is called an **inversion** layer. The source-drain channel behaviour somewhat like a **resistor**, and hence it is called the linear region. Analogue integrated circuits normally operate transistors in this linear region.

Saturation or Pinch off: If the drain voltage is now increased to that of the gate voltage, i.e.$V_{gd} < V_t$, the inversion layer underneath the gate at the drain terminal can no longer maintain the inversion layer. The channel now does not reach the drain. The channel is now being cut off (pinch off) as shown. However, conduction is still brought about by the drift of electrons under the influence of the positive drain voltage. The current is now independent of the drain voltage. This is the saturation state.

In digital VSLI, MOS transistors operate in either the cutoff or saturation state, only transit through the linear state during switching.

nMOS I-V characteristics



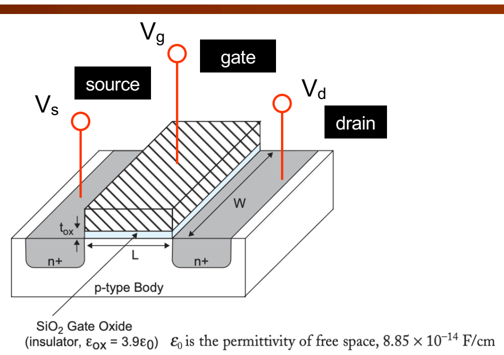
This graph shows the drain-source current vs drain-source voltage.

In the linear region, which is to the left of the dotted red line, the graphs shows more or less a straight line behaviour. The gradient (current/voltage) is the conductance (1/resistance) of the channel. This is dependent on on Vgs.

However, when Vgs < Vt, no current flows. This is the cutoff stage.

To the right of the dotted red line is where Vgd < Vt, saturation or pinch-off occurs because Vd is too high to maintain the inversion layer. Here Ids is effectively constant (flat in the characteristic) and is only dependent on Vgs and NOT Vds.

Key Transistor Equations for digital



Key current equation

$$I_{ds} = \begin{cases} 0 & V_{gs} < V_t & \text{Cutoff} \\ \cancel{\beta\left(V_{GT} - V_{ds}/2\right)V_{ds}} & \cancel{V_{ds} < V_{\text{dsat}}} & \cancel{\text{Linear}} \\ \dfrac{\beta}{2}V_{GT}^2 & V_{ds} > V_{\text{dsat}} & \text{Saturation} \end{cases}$$

$$\beta = \mu C_{ox}\frac{W}{L}$$

$$V_{GT} = V_{gs} - V_t$$

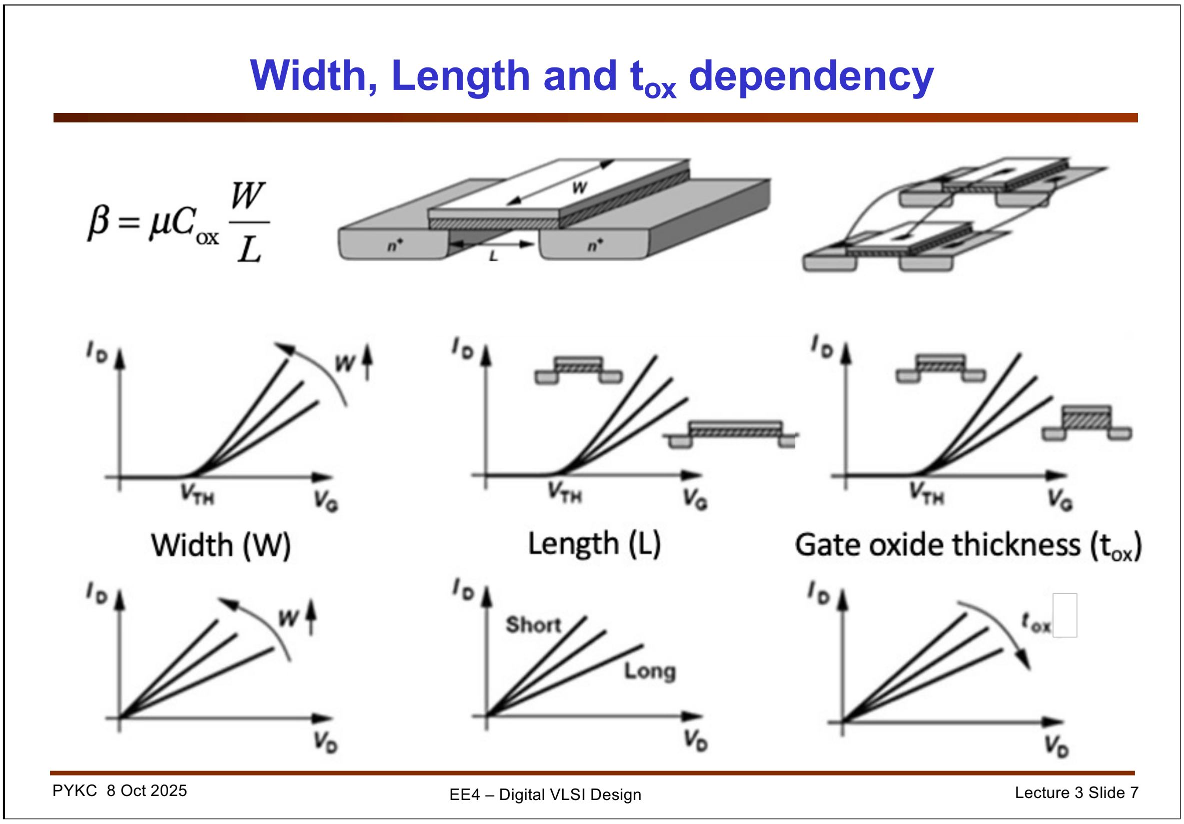
W&H p65-68

For digital CMOS design, the basic equation is simple.

The steady state condition of all digital transistors are either in cutoff or saturation state. Therefore, the current equation for Ids is either 0 if Vgs < Vt, or given by:

$$I_{ds} = \mu C_{ox}\frac{W}{L}\left(V_{gs} - V_t\right)^2 = \beta\, V_{GT}^2.$$

C_{ox} is dependent on the dielectric constant of the SiO_2 gate.
μ is the mobility of the majority carrier, in this case electronics.
W is the width of the transistor.
L is the length of the transistor.
V_{GS} is the gate voltage above the threshold voltage.
β is the constant for a transistor with fixed W and L, and is the "gain" of the transistor. Higher the gain β, higher the channel current I_{ds}.



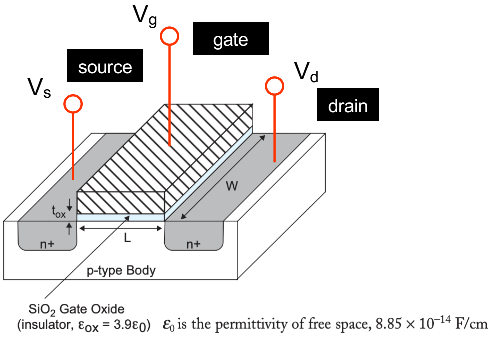
This slide shows how the channel current I_{ds} or I_d is affected by W, L and thickness of the oxide t_{ox}.

Larger the width, higher the gain and current.

Larger the length, lower the gain and current. Therefore in digital circuits, we normally only use the minimum possible length. For TSMC 65nm, the transistors are all 60nm in “drawn” length. (You will find out in Lab 2.)

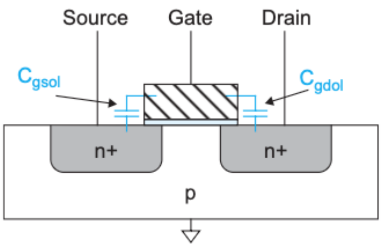
t_{ox} is the thickness of the gate oxide. The thinner it is, the higher the current because the electrical field is stronger.

Capacitance Model



- Gate capacitance as parallel plate:

$$C_0 = k_{ox}\varepsilon_0 \frac{WL}{t_{ox}} = \varepsilon_{ox}\frac{WL}{t_{ox}} = C_{ox}WL$$



Parameter	Cutoff	Linear	Saturation
C_{gb}	$\leq C_0$	0	0
C_{gs}	0	$C_0/2$	$2/3\ C_0$
C_{gd}	0	$C_0/2$	0
$C_g = C_{gs} + C_{gd} + C_{gb}$	C_0	C_0	$2/3\ C_0$

W&H p71-73

The capacitance seen at the gate terminal, in the simplest form, could be considered as a parallel plate with area = W x L between the gate metal and the substrate or the p-type body. Therefore the equation above provides a first-order approximation to the gate capacitance.

However, the situation is more complicated than this simple model.

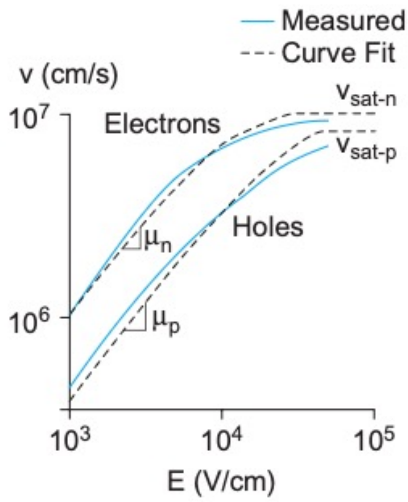
1. The gate is also connected to the source and drain. Therefore the capacitance seen at the gate is the sum of all three.
2. The source and substrate (body or bulk) are fixed, but the gate and drain voltages are changing during switching.

The table above shows the approximated effective capacitance at the gate for different state of the transistor.

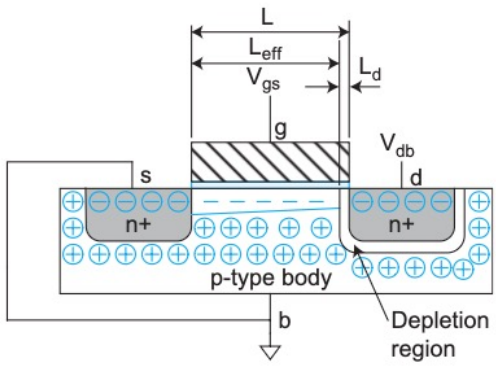
Why should you know this? It turns out that the delay of a digital gate is mainly governed by the drain current I_{ds} and the input capacitances seen by the output of the driving gate.

Non-ideal Effects (1) – on transistor gain

Mobility Degradation and Velocity Saturation



Channel Length Modulation



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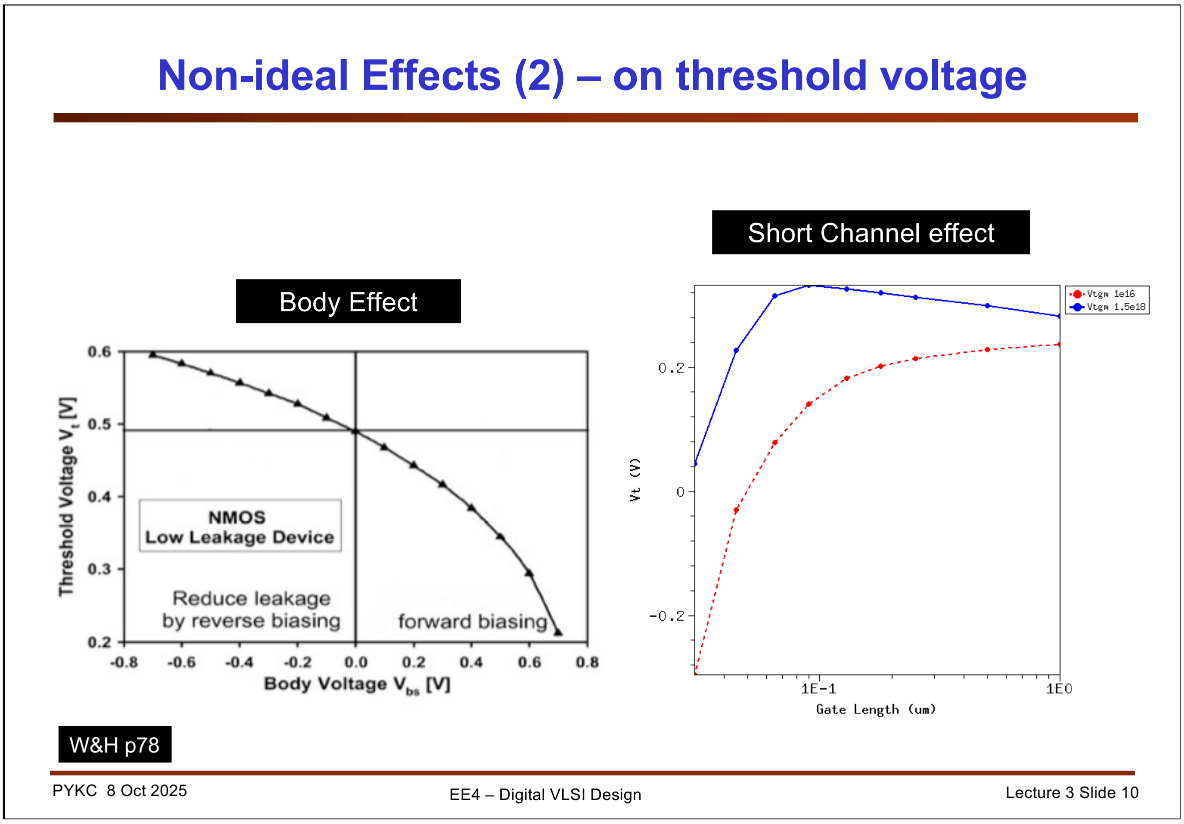
So far, we only consider simplified model of a MOS transistor. Actual transistors have other second or high-order effects that influence their behaviour.

1. **Mobility degradation**: I_{ds} is proportional to the mobility of the transistor μ. However, the mobility is dependent on the applied electric field, which is a proportional to V_{ds} for a given length. The graph above shows how the drift velocity (directly proportional to mobility) for holes and electronics changes with electric field.

2. **Velocity Saturation:** As the electrical field increase above a certain value, the drift velocity saturates at a certain level.

3. **Channel length modulation**: The p-n junction between the drain and the substrate form a reverse biased diode and depletion region is form. The thickness of this is dependent on the drain voltage. Therefore the effective channel length L_{eff} is dependent on the drain to body voltage. This effect is known as channel length modulation (i.e. the channel length is modulated by the drain voltage).

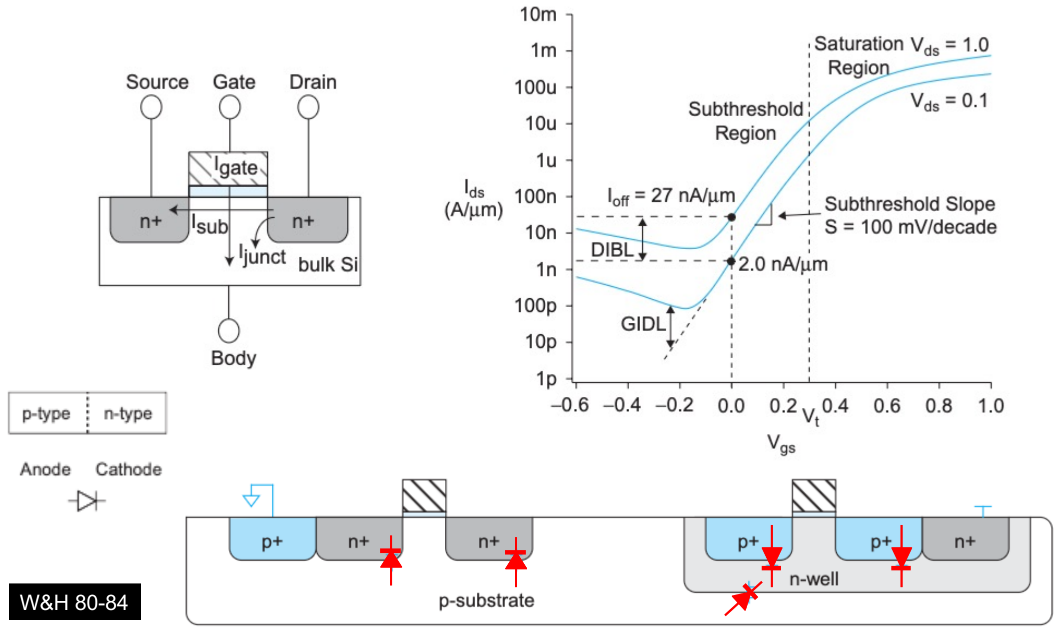


Here are more non-ideal effects, but now affecting threshold voltage of the transistor.

4. **Body effect**: We usually consider a transistor as a 3-terminal device. In reality it is a four terminal device with the fourth terminal being the body (or bulk) of the transistor. The threshold voltage is dependent on this voltage between the source and the body as shown above.

5. **Short channel effect**: As the gate length decreases, the threshold voltage Vt. typically decreases instead of staying constant.

Non-ideal Effects (3) - leakages



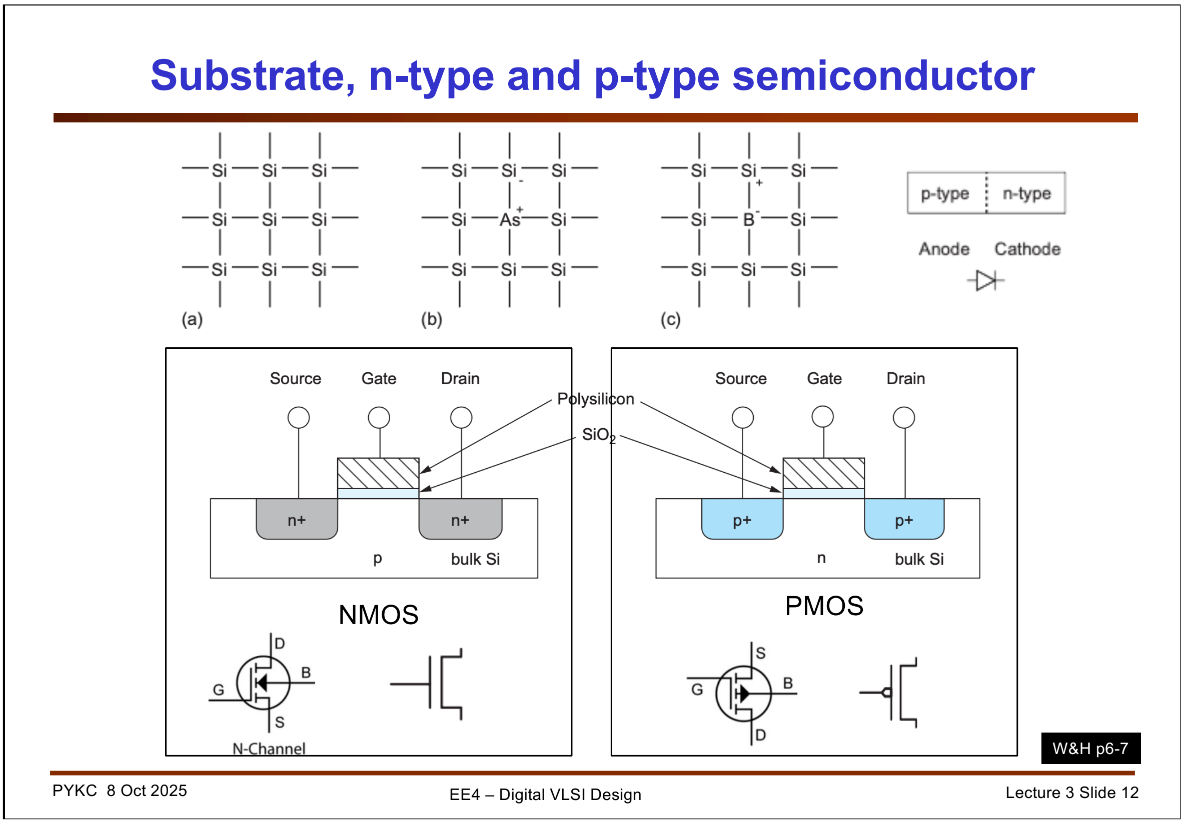
When a transistor is off, so far we assume that no current flows anywhere. This is not true. There are various leakage current happening:

- from the gate to body through the gate insulation
- across the reverse biased p-n junction between source/drain and the body, and
- across the drain to the source.

When a transistor is off, the gate voltage is below threshold voltage Vt, there is still current flow through the channel as shown in the graph above. Note that the y-axis is in log scale. This region is known as subthreshold region and the transistor is in the **weak inversion** state.

The diagram below show that p-n junction diodes found in CMOS transistors. They all have leakage currents in spite of the diode supposedly being not conducting.

Note that all these leakage currents are strongly dependent on the operating temperature of the transistor.



The next section is about the CMOS fabrication process.

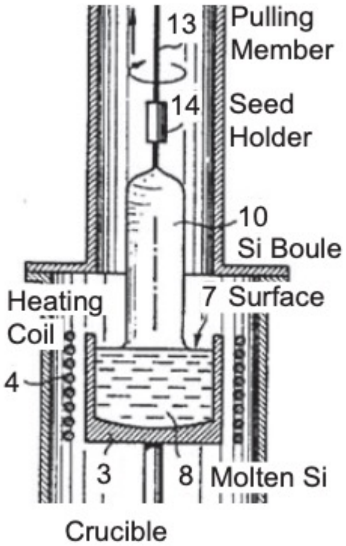
The substrate (or starting material) for chip is silicon. Impurity atoms are introduced to the silicon lattice through a process known as doping.

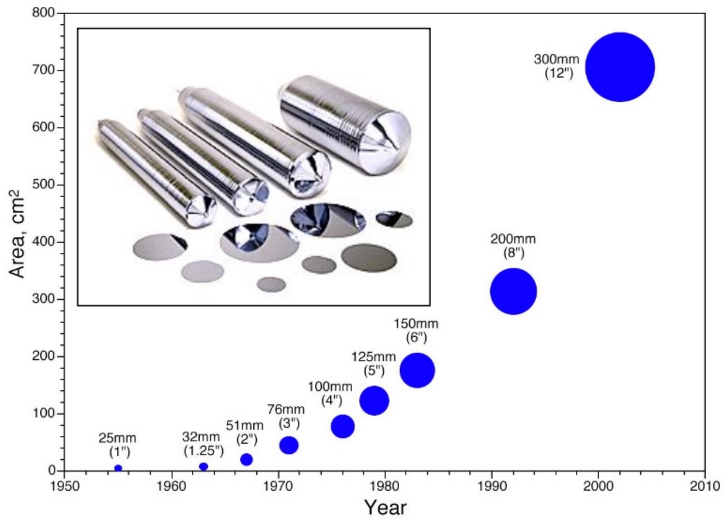
When silicon, which has four electrons, is doped with a dopant such as arsenic (middle figure), which has one extra electron which is weakly bonded and can come free under thermal vibration. The free electron can carry current, improving the conductivity of the doped silicon. This is n-type semiconductor (electrons have negative charge).

When silicon is doped with impurities with three electrons, such as Boron, it has a vacant space of an electron from the neighbouring silicon atom. When it “borrows” an electron from its silicon neighbour, the silicon atom is now one electron short, and has to borrow one from its own neighbour and so on. In this way electrons also flow, but to fill the vacant space. These vacant spaces are called holes.

In short, n-type material is doped with impurities with extra electron, which conduct electricity that way. P-type material is doped with impurities that has space (or a hole) for someone else donated electron. The carriers for n-type material are electrons. The carriers for p-type material are holes.

Silicon wafers



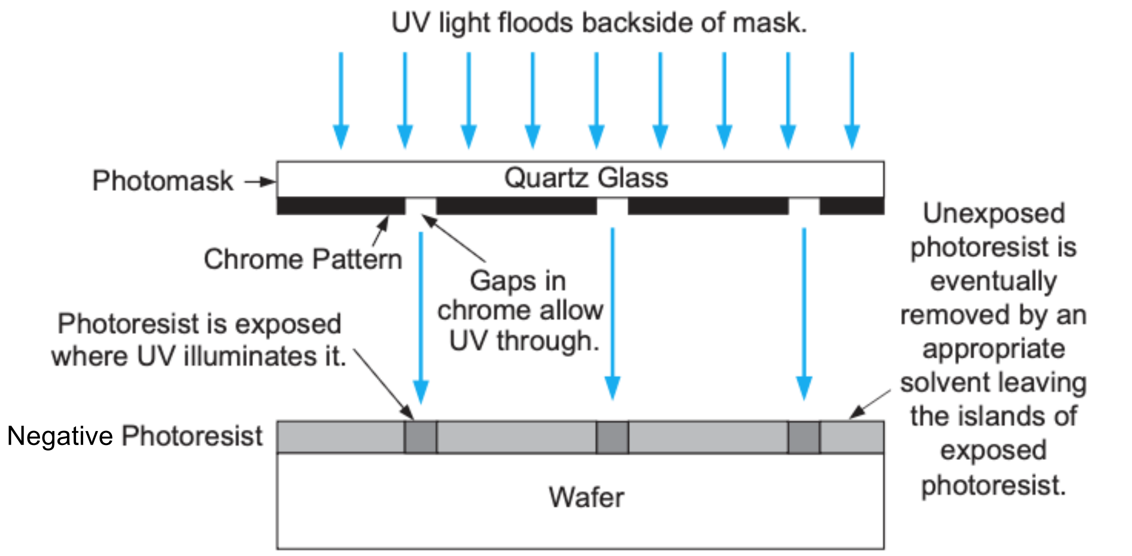


VLSI chips are manufactured on silicon wafers.

Wafers are cut from cylindrical ingots of single-crystal silicon, called boules. A boules is pulled from a crucible of pulled from a crucible of pure molten silicon. A seed crystal of silicon is dipped into the pool of molton silicon. The seed is gradually withdrawn and rotated. The diameter of the ingot produced is dependent on the rates of withdrawal and rotation.

The ingot is then cut into thin slices of silicon substrate wafers up to a diameter of 300mm.

Photolithography



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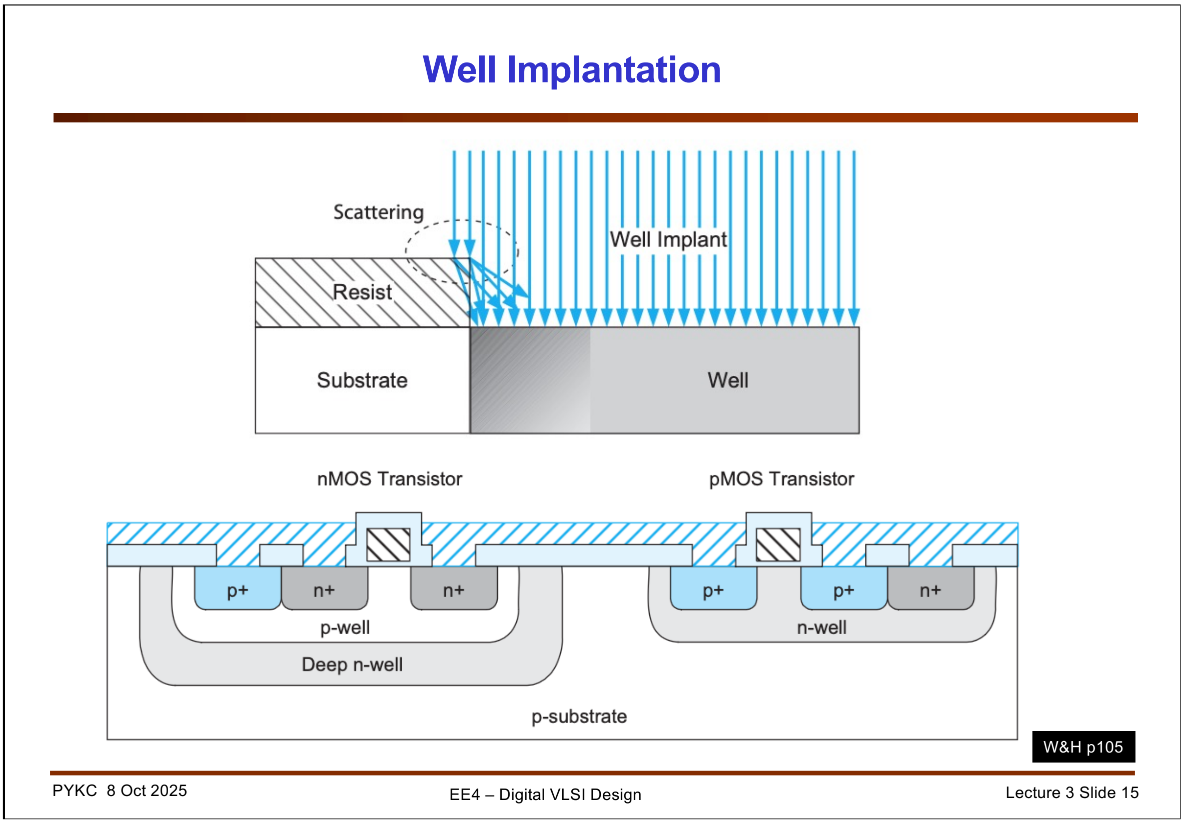
Photolithography uses a photomask with pattern containing gaps through which UV light can go through. The wafer is coated with a layer of photoresist which reacts to the UV light. Photoresist can be positive or negative.

Positive photoresist is initially insoluble to developer solvent unless it is exposed to the UV light. The gaps in the pattern mask determine the area that will be exposed.

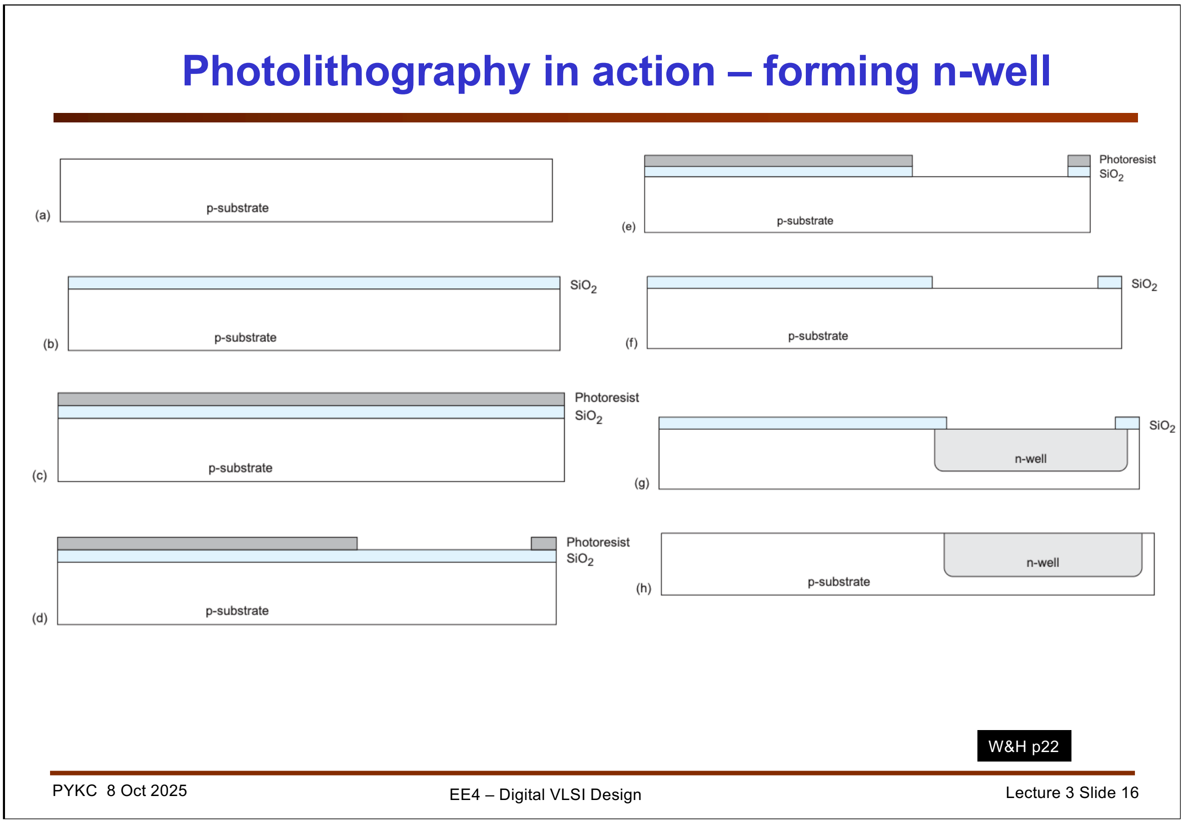
Negative photoresist is the opposite. It is soluble to solvent unless it is exposed. Therefore the gaps in the pattern determine islands of insoluble regions.



The areas where photoresist remain shield the substrate from implanting dopants. Here the windows into the layer of photoresist exposes the substrate area to an atmosphere of dopant molecules which diffuses into the substrate.

If an n-well is required, n-type impurities (e.g. arsenic) is implanted into the substrate to form n-well regions.

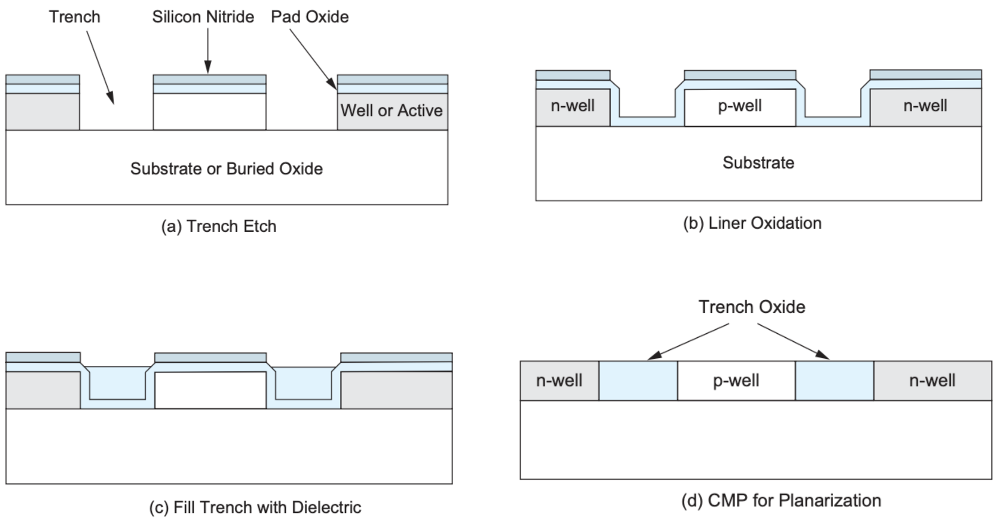
In some process, three wells are formed into which nMOS and pMOS transistors are fabricated as shown in the cross-section diagram above.



The actual step in using the photomask to form an n-well region require a number of steps shown above.

a) Start with a p-substrate of a lightly doped wafer.
b) Grow a layer of SiO_2 insulation by exposing the wafer to an atmosphere of oxygen at high temperature.
c) Spin a layer of positive photoresist.
d) UV light through hole in pattern on the mask defines the area where the n-well goes. The photoresist is dissolved in places that are exposed to UV. A window is opened up to the SiO_2 insulation.
e) The oxide layer is etched away (with acid) where it is not covered by photoresist, now opening up a window into the p-type semiconductor substrate.
f) The photoresist is removed by different type of acid.
g) Dopants are either diffused or implanted into the open area with the oxide layer providing a barrier. Not that with diffusion, the n-well region is larger than the window because diffusion happens in both the vertical and the lateral directions. This is called lateral diffusion.
h) The remaining layer oxide is removed with acid, leaving the original substrate with the added n-well region.

Shallow Trench Isolation (STI)



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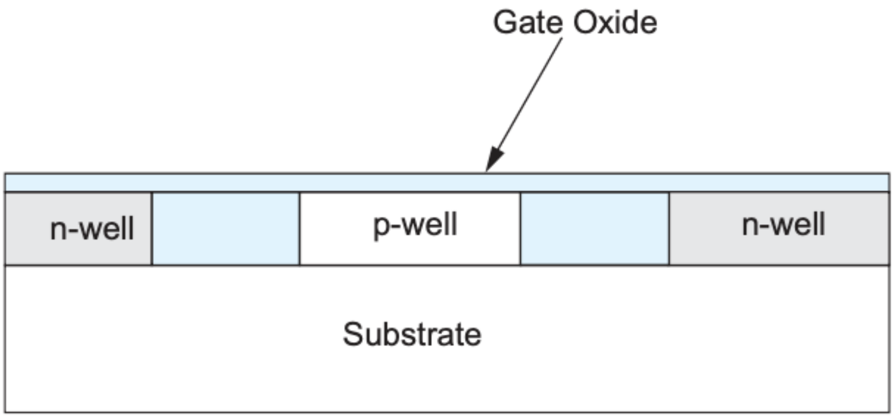
Shallow trench isolation (STI) is a process step that isolates individual devices from each other. It is used for 350nm process and under.

In between n-well and p-well region (where p- and n-type transistors are made respectively), small trenches are opened up in the substrate. A thin layer of liner oxide is grown to over the open trenches. Then the trenches are filled with SiO_2. The temporary oxide layers are removed using a Chemical Mechanical Polishing (CMP) step.

Gate Oxide Layer

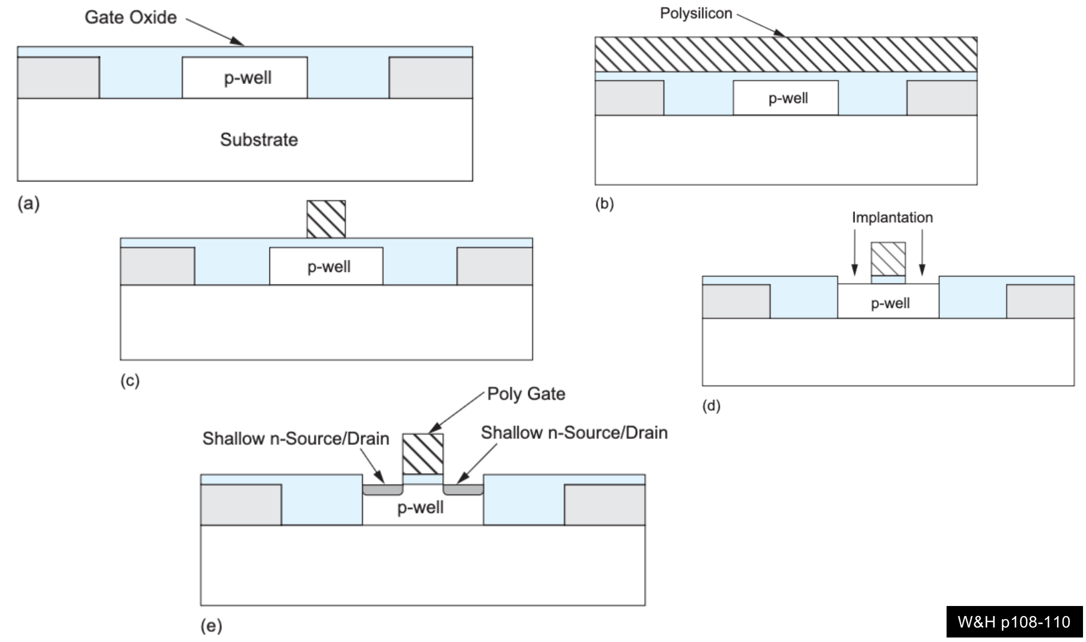


W&H p107

So far, we have shown how n-well and p-well are formed. Into these, we make p-type and n-type transistors respectively.

The next step is to grow a very thin and uniform SiO_2 layer that forms the gate oxide. For a 65nm process node, the effective thickness of the gate oxide is 10.5 to 5 angstrom (10^{-10}m).

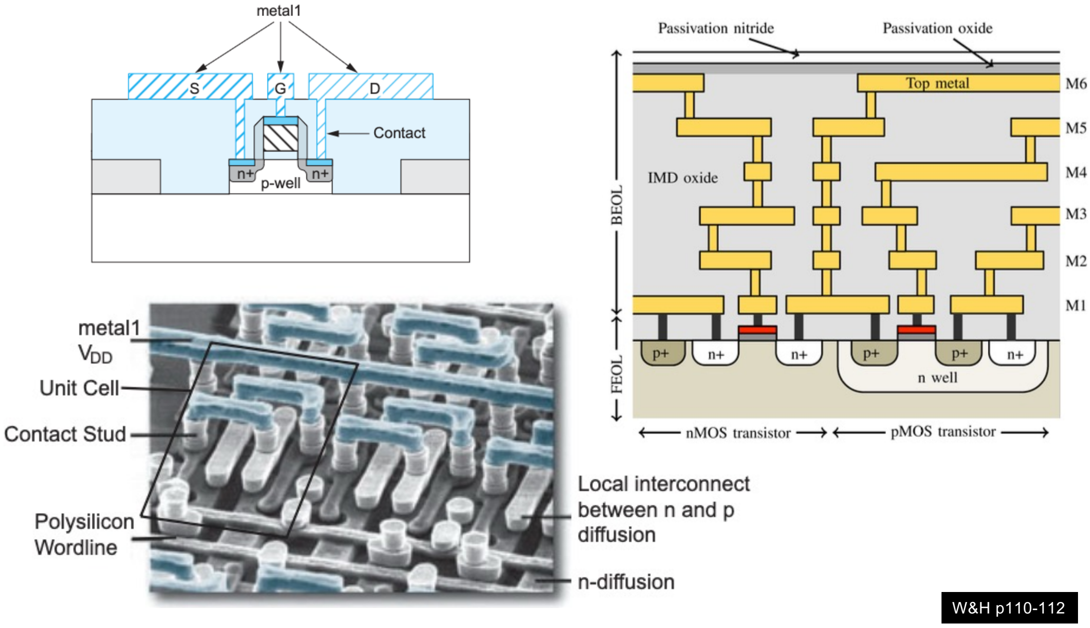
Gate and Source/Drain Formation



When silicon is deposited on SiO2 without crystal orientation, it forms polycrystalline silicon, commonly called polysilicon or simply poly. This is the material that forms the gate of the transistor.

a) A thin layer of SiO2 is grown onto the trench wafer with n and p well formed.
b) A layer of polysilicon is deposited on the thin gate oxide.
c) Photolithography technique is used to remove the polysilicon except for the gates of transistors.
d) The polysilicon gate is now forming the mask for the implantation that forms the source and drain of the transistor. This guarantee that the gate region is perfectly aligned. It is therefore called a *self-alighed polysilicon gate* process.
e) The end result is a n-type transistor.

Contacts and Metalization



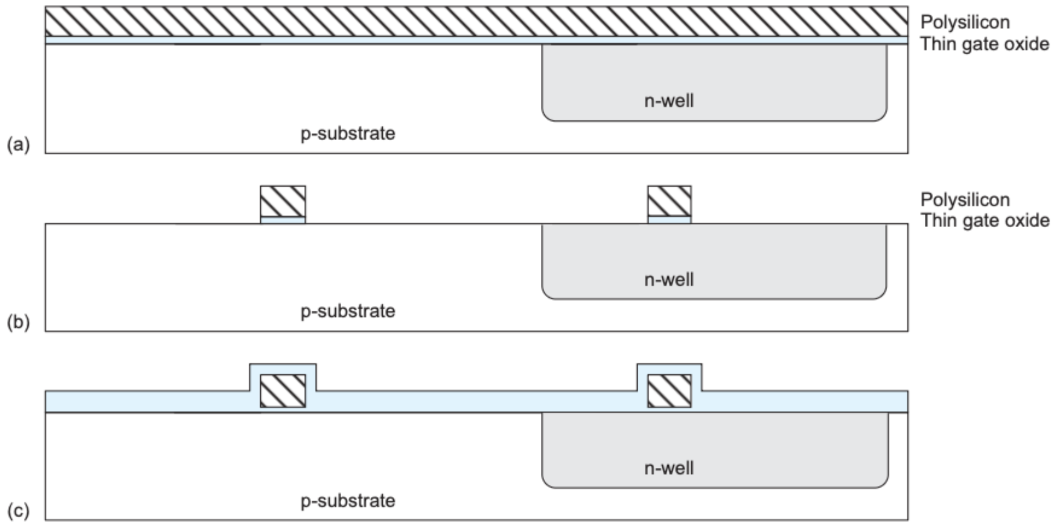
Contact are holes etched in the dielectric or oxide layer after the sources and drains are formed. Then metal 1 layer formed by aluminum is added by either evaporation or sputtering process.

Evaporation is performed by passing a high electrical current through a thick aluminum wire in a vacuum chamber. Some of the aluminum atoms are vaporized and deposited on the wafer.

Sputtering is achieved by generating a gas plasma by ionizing an inert gas using an RF or DC electric field. The ions are focused on an aluminum target and the plasma dislodges metal atoms, which are then deposited on the wafer.

The photograph shows the different interconnection layers found in a SRAM chip.

Putting all steps together (1)



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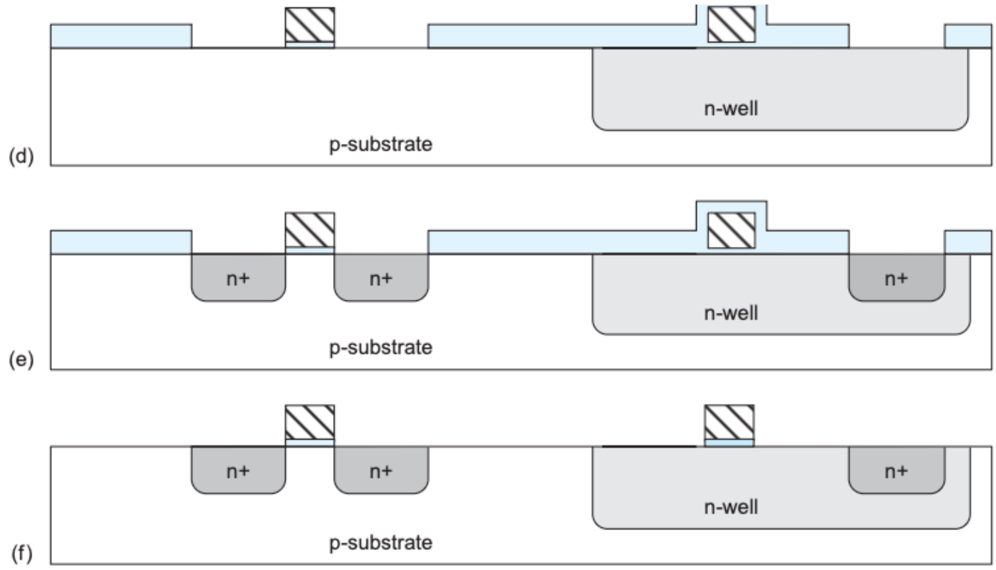
The next three slides shows the steps in making an n-type and a p-type transistor to form an inverter.

Putting all steps together (2)



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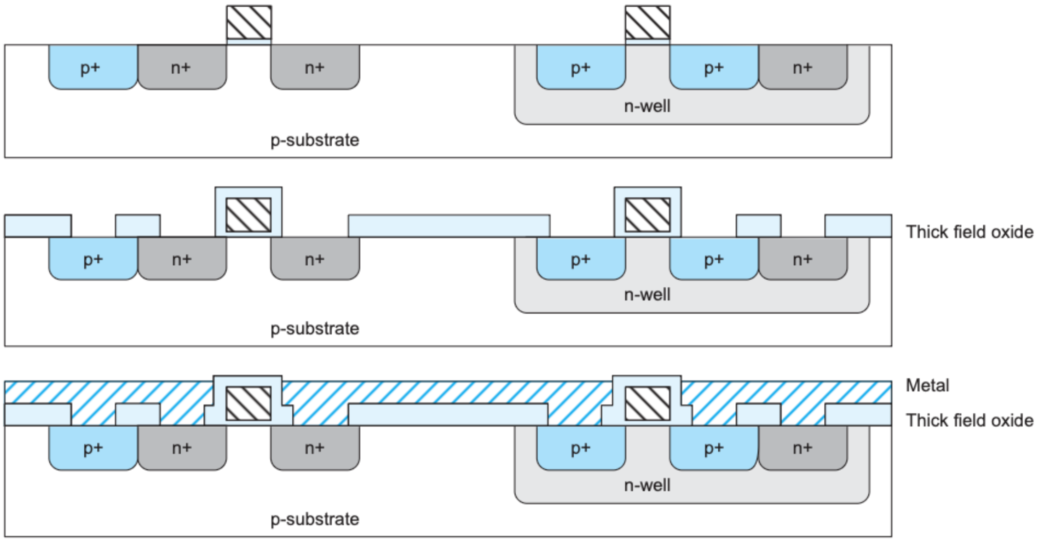
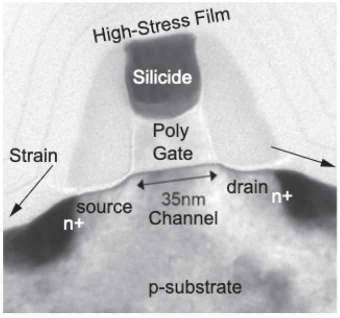
Putting all steps together (3)

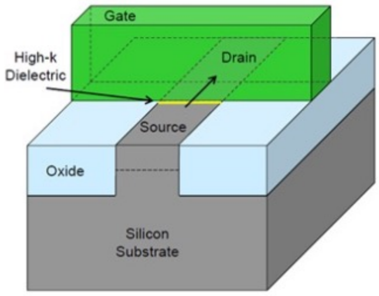


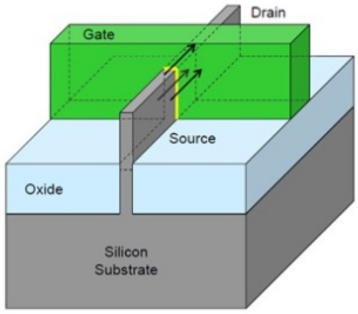
W&H p23-24

Process Enhancements

- **Trench isolation (L3S17):** Shallow trench isolation (STI) prevents electrical current leakage between adjacent semiconductor device components.
- **High-K dielectrics/metal gate**: Replacing the SiO2 gate dielectric with a high-K material allows increased gate capacitance without the associated leakage effects.
- **Strained silicon**: A layer of silicon in which the silicon atoms are stretched beyond their normal interatomic distance leading to better mobility, resulting in better chip performance and lowering energy consumption.
- **Gate engineering**: for within-die choice of multiple transistor threshold voltages (Vt) to optimize delay or power.
  - 3D MOSFET structure (FINFET) to improve controllability and reduce leakage.







Here are four ways that the chip fabrication process has been enhanced in recent years.

Further Exploration

- Read the relevant part of Weste and Harris' book, part of Chapter 2 and 3.
- "How are chips made?" https://www.youtube.com/watch?v=IkRXpFIRUl4
- "Photolithography" https://www.youtube.com/watch?v=TjpbrLTem3M
- "TSMC Fab Tour" https://www.youtube.com/watch?v=divOKxuYklM

To gain further understanding, I recommend you to read the sections (pages) of the Weste and Harris textbook indicated on each slide.

Here are three YouTube videos that you may find information about the silicon fabrication process.